# Omar AbedelKader

PhD in Computer Science
EVREF - Inria / Progress - LaBRI
University of Lille / University of Bordeaux
Lille, France

Updated: October 28, 2026
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**Interests:** Artificial Intelligence · Software Engineering · LLMs · AI4SE

## About Me

*Biography:*

- I am an AI researcher with extensive experience in artificial intelligence and software engineering.
- I am currently pursuing a Ph.D. in Computer Science at the University of Lille.
- I am currently involved in several open-source, research, AI, and community initiatives, where I contribute as a founder, maintainer, project leader, president, and active member.
- You can see the full list of my current responsibilities **here**.

*Awards:*

- Best Paper Award (3rd place) — IWST 2025, Gdańsk, Poland
- Alumni Testimonial — IDMC 2024, University of Lorraine, France *

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## Publications

*Rank A*

- Kier, K., Giagnorio, A., Abedelkader, O., Zaitsev, O., Peharz, R., Robbes, R., Bavota, G., & Ducasse, S. (2026). Teaching LLMs a Low-Resource Language: Enhancing Code Completion in Pharo. *International Conference on Software Maintenance and Evolution (ICSME)*. DOI.

*Rank C*

- AbedelKader, O., Ducasse, S., Zaitsev, O., Robbes, R., & Polito, G. (2025). Package-Aware Approach for Repository-Level Code Completion in Pharo. *International Workshop on Smalltalk Technologies (IWST)*. DOI.
- Elkolei, M., AbedelKader, O., Ducasse, S., & Zaitsev (2026). Alternative UX Extensions and Their Trade-offs for Code Completion in Pharo. *International Workshop on Smalltalk Technologies (IWST)*. DOI.

*Q2 Journals*

- AbedelKader, O., Ducasse, S., Zaitsev, O., Robbes, R., & Polito, G. (2026). Evaluating Package-Level Scoping Strategies for Repository-Level Code Completion in Pharo. *Journal of Computer Languages (COLA)*. DOI.

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## Services

*2027*

- Junior PC Member in Junior Program Committee at International Conference on Technical Debt'27
- Committee Member in Technical Papers - Junior Program Committee, Junior PC at Mining Software Repositories'27
- Committee Member in Artifact Track at FormaliSE'27
- Shadow Reviewer at ICSE'27
- Contributor at Café IA (Conseil National du Numérique).
- Member at Club IA et Transition écologique (Ministère de la Transition Écologique).
- Website Maintainer GDR-SciLog
- Project Leader and Maintainer for multiples projects (ChatPharo, Pharo-Infer and many more).
- Member in the organization Pharo-AI.
- Founder and Maintainer of the organization Pharo-LLM.

*2026*

- Member of the organizing committee, GDR-SciLog'26.
- Member of the organizing committee, ESUG'26.
- Contributor at Café IA (Conseil National du Numérique).
- Member at Club IA et Transition écologique (Ministère de la Transition Écologique).
- President of Synapse-NeuroTech-Lille, a student association focused on AI and neuroscience.
- Website Maintainer GDR-SciLog
- Project Leader and Maintainer for multiples projects (ChatPharo, Pharo-Infer and many more).
- Member in the organization Pharo-AI.
- Founder and Maintainer of the organization Pharo-LLM.

*2025*

- Member of the organizing committee ESUG'25
- Sub reviewer at BENEVOL'25.
- Contributor at Café IA (Conseil National du Numérique).
- Member at Club IA et Transition écologique (Ministère de la Transition Écologique).

- Project Leader and Maintainer for multiples projects (ChatPharo, Pharo-Infer and many more).
- Member in the organization Pharo-AI.
- Founder and Maintainer of the organization Pharo-LLM.

*2024*

- Member of the organizing committee ESUG'24.
- Contributor at Café IA (Conseil National du Numérique).
- Member at Club IA et Transition écologique (Ministère de la Transition Écologique).

*2023*

- Volunteer First Aider in the French Red Cross

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**Teaching**

*2027*

- Teaching assistant at IUT : *Software Maintenance* 12h Tutorials IUT-Lille
- Teaching assistant at Polytech: *DataBase* 16h Labs Polytech-Lille
- Teaching assistant at Polytech: *Internet* 12h Labs Polytech-Lille

*2026*

- Teaching assistant at IUT: *Introduction to Software Development* 41,5 h Labs IUT-Lille
- Teaching assistant at IUT : *Software Maintenance* 12h Tutorials IUT-Lille
- Teaching assistant at Polytech: *DataBase* 16h Labs Polytech-Lille
- Teaching assistant at Polytech: *Internet* 12h Labs Polytech-Lille

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**Software**

*ChatPharo*

- Live conversational interface between Pharo developers and large language models.
- Enables interactive discussions with LLMs directly from the Pharo environment.
- Designed an open and extensible architecture to experiment with multiple LLM backends.
- GitHub Repository: ChatPharo

*INERIS-IA*

- AI platform developed at INERIS for document classification and knowledge management.
- Built a Flask-based web interface integrating multiple ML and NLP models.
- Trained document classification models aligned with COP 2027 objectives and INERIS strategic themes.
- Implemented additional features including document similarity, keyword extraction, and corpus quality improvement.
- GitHub Repository: INERIS-IA

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**Students**

*2026*

- Mehdi Elkolei, Bachelor Intership, 4 month, April 2026 - July 2026
  I supervised Mehdi Elkolei on his work on extending Complishon, Pharo's context-aware code completion engine. His project focused on four UX-oriented extensions: typo tolerance, implicit prefix expansion, grouped completion entries, and camel-case matching. I guided the technical direction and design choices, with particular attention to usability, modularity, and integration with the existing architecture. Elkolei, M., AbedelKader, O., Ducasse, S., & Zaitsev (2026). Alternative UX Extensions and Their Trade-offs for Code Completion in Pharo. *International Workshop on Smalltalk Technologies (IWST)*. DOI.

*2025*

- Srajal Dwivedi, Google Summer of Code Intership, 3 month, May 2025 - July 2025 I supervised Srajal the ChatPharo project, which aimed to integrate LLMs into the Pharo IDE. It was a Google Summer of Code 2025 project. Srajal improved ChatPharo's user interface and implemented new features to facilitate interaction with LLMs. The project was well-received by the Pharo community and contributed to the adoption of ChatPharo by developers. It was also one of the first projects to integrate LLMs into the Pharo IDE.

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**Public Talks**

*2026:*

- ESUG 2026: ChatPharo: an AI Assistant inside Pharo
- ESUG 2026: Pharo-LLM: Building a sovereign LLM ecosystem in Pharo
- French Brain Awareness Week: AI and the brain: two ways of learning, what does that change for our daily lives?

*2025:*

- ESUG 2025: ChatPharo: An Open Architecture for Understanding How to Talk Live to LLMs, Gdansk, Poland

- GT GL: ChatPharo: Talking Live to LLMs from Pharo, Lille, France
- Scientific Seminar: LLM101-2, Lille, France
- Scientific Seminar: LLM101, Lille, France

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**Professional Experience**

- **PhD Student** · *Oct. 2024 - Oct. 2027*
  INRIA Lille / LaBRI Bordeaux
- **Invited Researcher** · *Sept. 2026 - Dec. 2026*
  Polytechnique Montréal · Montréal, Canada
- **AI Engineer** · *Sept. 2023 - Sept. 2024*
  Institut national de l'environnement industriel et des risques (INERIS) · Verneuil-en-Halatte, France
- **Intern** · *June 2023 - Aug. 2023*
  Laboratoire d'Informatique de Paris Nord (LIPN) · Villetaneuse, France

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**Education**

- **Ph.D. in Computer Science**
  University of Lille
- **M.Sc. in Natural Language Processing**
  University of Lorraine
- **B.Sc. in Data Science**
  Lebanese University

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**Hobbies**

- **Water Activities:** Swimming · Diving
- **Outdoor and Adventure Sports:** Hiking · Skiing · Skydiving
- **Creative and Artistic Activities:** Photography
- **Precision Sports:** Boxing · Bowling · Golf

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**Languages**

- **French:** Native
- **English:** B2

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